CLASS 9 · CBSE · GANITA MANJARI PART 1 · FORMULA BANK

Chapter 4: Exploring Algebraic Identities

Every formula and result of the chapter on one sheet

1	Square of a sum	$(a+b)^2 = a^2 + 2ab + b^2$
2	Square of a difference	$(a-b)^2 = a^2 - 2ab + b^2$
3	Square of three terms	$(a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
4	Difference of squares	$a^2 - b^2 = (a+b)(a-b)$
5	Quick squaring	$a^2 = (a+b)(a-b) + b^2$
6	Product of two binomials	$(x+a)(x+b) = x^2 + (a+b)x + ab$
7	General product	$(px+a)(qx+b) = pqx^2 + (pb+qa)x + ab$
8	Cube of a sum	$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
9	Cube of a difference	$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
10	Sum of cubes	$x^3 + y^3 = (x+y)(x^2 - xy + y^2)$
11	Difference of cubes	$x^3 - y^3 = (x-y)(x^2 + xy + y^2)$
12	Three cubes	$x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)$
13	When $x + y + z = 0$	$x^3 + y^3 + z^3 = 3xyz$
14	Splitting the middle term	For $x^2 + sx + p$: find two numbers with sum s and product p